Digital Electronics Question and Answers

Memory Devices-1

Topic 10

1.) Memory Devices

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Digital Electronics Question & Answers

1.) Memory Devices

1. Memory is a/an ___________
a) Device to collect data from other computer
b) Block of data to keep data separately
c) Indispensable part of computer
d) Device to connect through all over the world

Answer: c
Explanation: Memory is an indispensable unit of a computer and microprocessor based systems which stores permanent or temporary data.

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2. The instruction used in a program for executing them is stored in the ___________
a) CPU
b) Control Unit
c) Memory
d) Microprocessor
Answer: c
Explanation: All of the program and the instructions are stored in the memory. The processor fetches it as and when required.

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3. A flip flop stores ___________
a) 10 bit of information
b) 1 bit of information
c) 2 bit of information
d) 3-bit information
Answer: b
Explanation: A flip-flop has capability to store 1 bit of information. It can be used further after erasing previous information.

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4. A register is able to hold ___________
a) Data
b) Word
c) Nibble
d) Both data and word

Answer: b
Explanation: Register is also a part of memory inside a computer. It stands there to hold a word. A word is a group of 16-bits or 2-bytes.

5. A register file holds ___________
a) A large number of word of information
b) A small number of word of information
c) A large number of programs
d) A modest number of words of information
Answer: d
Explanation: A register file is different from a simple register because of capability to hold a modest number of words of information. A word is a group of 16-bits or 2-bytes.

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6. The very first computer memory consisted of ___________
a) A small display
b) A large memory storage equipment
c) An automatic keyboard input
d) An automatic mouse input

Answer: b
Explanation: The very first computer memory consisted of a minute magnetic toroid, which required large, bulky circuit boards stored in large cabinates.

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7. A minute magnetic toroid is also called as ___________
a) Large memory
b) Small memory
c) Core memory
d) Both small and large memory

Answer: c
Explanation: A minute magnetic toroid is also called as core memory which is made up of a semiconductor. A semiconductor is a device whose electrical conductivity lies between that of conductor and insulator.

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8. Which one of the following has capability to store data in extremely high densities?
a) Register
b) Capacitor
c) Semiconductor
d) Flip-Flop
Answer: c
Explanation: Semiconductor has capability to store data in extremely high densities.

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9. A large memory is compressed into a small one by using ___________
a) LSI semiconductor
b) VLSI semiconductor

c) CDR semiconductor
d) SSI semiconductor

Answer: b
Explanation: VLSI (Very Large Scale Integration) semiconductor is used in modern computers to short the size of memory.

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10. VLSI chip utilizes ___________
a) NMOS
b) CMOS
c) BJT
d) All of the Mentioned
Answer: d
Explanation: VLSI (Very Large Scale Integration) is a memory chip which is made up of NMOS, CMOS, BJT, and BiCMOS. It can include 10,000 to 100,000 gates per IC.

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11. CD-ROM refers to ___________
a) Floppy disk
b) Compact Disk-Read Only Memory
c) Compressed Disk-Read Only Memory
d) Compressed Disk- Random Access Memory

Answer: b
Explanation: CD-ROM refers to Compact Disk-Read Only Memory.

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12. Data stored in an electronic memory cell can be accessed at random and on demand using ___________
a) Memory addressing
b) Direct addressing
c) Indirect addressing
d) Control Unit
Answer: b
Explanation: Direct addressing eliminates the need to process a large stream of irrelevant data in order to the desired data word.

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13. The full form of PLD is ___________
a) Programmable Large Device
b) Programmable Long Device
c) Programmable Logic Device
d) Programmable Lengthy Device

Answer: c

Explanation: The full form of PLD is Programmable Logic Device.

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14. The evolution of PLD began with ___________

a) EROM

b) RAM

c) PROM

d) EEPROM

Answer: c

Explanation: The evolution of PLD (Programmable Logic Device) began with Programmable Read Only Memory (i.e. PROM). Here, the ROM can be externally programmed as per the user.

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15. A ROM is defined as ___________

a) Read Out Memory

b) Read Once Memory

c) Read Only Memory

d) Read One Memory

Answer: c

Explanation: A ROM is defined as Read Only Memory which can read the instruction stored in a computer.

16.The full form of ROM is ___________

a) Read Outside Memory

b) Read Out Memory

c) Read Only Memory

d) Read One Memory

Answer: c

Explanation: The full form of ROM is Read Only Memory.

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17. ROM consist of ___________

a) NOR and OR arrays

b) NAND and NOR arrays

c) NAND and OR arrays

d) NOR and AND arrays

Answer: c

Explanation: ROM consists of NAND and OR arrays which can be programmed by the user to implement combinational & sequential functions. Combinational Operations like that of adders and subtractors and Sequential Functions like that of storing in the memory.

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18. For reprogrammability, PLDs use ___________
a) PROM
b) EPROM
c) CDROM
d) PLA

Answer: b

Explanation: For reprogrammability, PLDs use EPROM (i.e. Erasable PROM). It erases the previous program and starts uploading a new one. However, data is erased by exposing it to UV-light, which is a tedious and time-consuming process.

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19. The full form of PROM is ___________
a) Previous Read Only Memory
b) Programmable Read Out Memory
c) Programmable Read Only Memory
d) Previous Read Out Memory

Answer: c

Explanation: The full form of PROM is Programmable Read Only Memory, where the ROM can be programmed by the user.

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20. The full form of EPROM is ___________
a) Easy Programmable Read Only Memory
b) Erasable Programmable Read Only Memory
c) Eradicate Programmable Read Only Memory
d) Easy Programmable Read Out Memory

Answer: b

Explanation: The full form of EPROM is Erasable Programmable Read Only Memory, where the ROM can be erased and re-used by the user.

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21. PLDs with programmable AND and fixed OR arrays are called ___________
a) PAL
b) PLA
c) APL
d) PPL

Answer: a

Explanation: PLDs with programmable AND and fixed OR arrays are called PAL (i.e. Programmable Array Logic). However, PAL is less flexible but has higher speed.

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22. When both the AND and OR are programmable, such PLDs are known as ___________
a) PAL
b) PPL

c) PLA
d) APL
Answer: c
Explanation: When both the AND and OR are programmable, such PLDs are known as PLA (i.e. Programmable Logic Array). However, PLA is more flexible but has less speed.

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23. ASIC stands for __________
a) Application Special Integrated Circuits
b) Applied Special Integrated Circuits
c) Application Specific Integrated Circuits
d) Applied Specific Integrated Circuits
Answer: c
Explanation: In digital electronics, ASIC stands for Application Specific Integrated Circuits. It is a customized integrated circuit which is produced for a specific use and not for a common-purpose.

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24. The programmability and high density of PLDs make them useful in the design of __________
a) ISAC
b) ASIC
c) SACC
d) CISF

Answer: b
Explanation: The programmability and high density of PLDs make them useful in the design of ASIC (i.e. Application Specific Integrated Circuits) where design changes can be more rapidly and inexpensively.

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25. FPGA stands for __________
a) Full Programmable Gate Array
b) Full Programmable Genuine Array
c) First Programmable Gate Array
d) Field Programmable Gate Array
Answer: d
Explanation: In digital electronics, FPGA stands for Field Programmable Gate Array. This type of integrated circuit is for general-purpose which is configured by the user as per their requirement.

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26. Which of the following is a reprogrammable gate array?
a) EPROM
b) FPGA

c) Both EPROM and FPGA
d) ROM
Answer: c
Explanation: Both FPGA and EPROM are reprogrammable gate array.

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27. The difference between FPGA and PLD is that ___________
a) FPGA is slower than PLD
b) FPGA has high power dissipation
c) FPGA incorporates logic blocks
d) All of the Mentioned
Answer: c
Explanation: The difference between FPGA and PLD is that FPGA incorporates logic blocks instead of fixed AND-OR gates and is faster with low power dissipation. FPGAs are designed for having higher gate count whereas, PLDs are used for lesser gate counts.

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28. Memories are classified into ______ categories.
a) 3
b) 4
c) 5
d) 6
Answer: c
Explanation: Memory is typically classified of 2 types: Primary and Secondary. These are further classified into 5 types of memories and these are Secondary, RAM, Dynamic/Static, Volatile/Non-volatile, Magnetic/Semiconductor Memory.

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29. Secondary memory is also known as ____________
a) Registers
b) Main Memory
c) RAM
d) Both registers and main memory
Answer: d
Explanation: Secondary memory is also known as Registers/Main Memory. In secondary memory, data is usually stored for a long-term.

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30. In a computer, registers are present ___________
a) Within control unit
b) Within RAM
c) Within ROM
d) Within CPU
Answer: d

Explanation: In a computer, registers are present within the CPU to store data temporarily during arithmetic and logical operations and during the functioning of the ALU.

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31. Which of the following has the lowest access time?
a) RAM
b) ROM
c) Registers
d) Flag
Answer: c
Explanation: Registers has the lowest access time, as they are available inside the CPU. Registers are present within the CPU to store data temporarily during arithmetic and logical operations and during the functioning of the ALU.

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35. Main memories of a computer, usually made up of ___________
a) Registers
b) Semiconductors
c) Counters
d) PLDs
Answer: b
Explanation: Main memories of a computer, usually made up of semiconductors which are available external to the CPU to store program and data during execution of a program. Registers are present within the CPU to store data temporarily during arithmetic and logical operations and during the functioning of the ALU.

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36. As the storage capacity of the main memory is inadequate, which memory is used to enhance it?
a) Secondary Memory
b) Auxiliary Memory
c) Static Memory
d) Both Secondary Memory and Auxiliary Memory

Answer: d
Explanation: As the storage capacity of the main memory is inadequate, Secondary memory is used to enhance it and it is also known as auxiliary memory. Secondary memory is also known as Registers/Main Memory. In secondary memory, data is usually stored for a long-term.

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37. Which memories are if magnetic memory type?
a) Main Memory
b) Secondary Memory

c) Static Memory
d) Volatile Memory

Answer: b
Explanation: Usually, secondary memories are of magnetic memory type that are used to store large type quantities of data. In secondary memory, data is usually stored for a long-term.

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38. Which of the following comes under secondary memory/ies?
a) Floppy disk
b) Magnetic drum
c) Hard disk
d) All of the Mentioned
Answer: d
Explanation: All of the mentioned equipments are of external storage which is known as secondary memories. In secondary memory, data is usually stored for a long-term.

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39. Based on method of access, memory devices are classified into ____________ categories.
a) 2
b) 3
c) 4
d) 5

Answer: a
Explanation: Based on the method of access, memory devices are classified into two categories and these are sequential access memory and RAM. A sequential access memory is one in which a particular memory location is accessed sequentially.

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40. A sequential access memory is one in which ___________
a) A particular memory location is accessed rapidly
b) A particular memory location is accessed sequentially
c) A particular memory location is accessed serially
d) A particular memory location is accessed parallely
Answer: b
Explanation: A sequential access memory is one in which A particular memory location is accessed sequentially (i.e. the ith memory location is accessed only after sequencing through previous (i-1) memory locations).

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41. An example of sequential access memory is ___________
a) Floppy disk

b) Hard disk
c) Magnetic tape memory
d) RAM

Answer: c
Explanation: A sequential access memory is one in which a particular memory location is accessed sequentially. In magnetic tape memory, data is accessed sequentially.

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42. A Random Access Memory is one in which __________
a) Any location can be accessed sequentially
b) Any location can be accessed randomly
c) Any location can be accessed serially
d) Any location can be accessed parallely
Answer: b
Explanation: A Random Access Memory is one in which any location can be accessed randomly.

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43. An example of RAM is __________
a) Floppy disk
b) Hard disk
c) Magnetic tape memory
d) Semiconductor RAM
Answer: d
Explanation: A Random Access Memory is one in which any location can be accessed randomly. A semiconductor RAM is too much fast and can occupy any space in the memory location.

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44. A static memory is one in which __________
a) Content changes with time
b) Content doesn't changes with time
c) Memory is static always
d) Memory is dynamic always
Answer: b
Explanation: A static memory is one in which content doesn't changes with time (i.e. stable). Dynamic memory is one in which content changes with time (i.e. unstable).

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45. A dynamic memory is one in which __________
a) Content changes with time
b) Content doesn't changes with time
c) Memory is static always
d) Memory is dynamic always

Answer: a

Explanation: A static memory is one in which content doesn't change with time (i.e. stable). Dynamic memory is one in which content changes with time (i.e. unstable).

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46. Dynamic memory cells use ________________ as the storage device.
a) The reactance of a transistor
b) The impedance of a transistor
c) The capacitance of a transistor
d) The inductance of a transistor

Answer: c

Explanation: Capacitance of a transistor prevents from loss of information in a dynamic memory cell.

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47. To store 1-bit of information, how many transistor is/are used _____________
a) 1
b) 2
c) 3
d) 4

Answer: a

Explanation: Only one bit transistor is needed to store 1-bit of information.

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48. Static memory holds data as long as ___________
a) AC power is applied
b) DC power is applied
c) Capacitor is fully charged
d) High Conductivity

Answer: b

Explanation: In any semiconductor equipment, AC power can't be supplied directly. So, static memory holds the data as long as DC power is applied.

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49. The example of dynamic memory is ___________
a) CCD
b) Semiconductor dynamic RAM
c) Both CCD and semiconductor dynamic RAM
d) Floppy-Disk

Answer: c

Explanation: The examples of dynamic memories are CCD and semiconductor dynamic RAM because of the contents of both the memories changes with time.

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50. In dynamic memory, CCD stands for __________
a) Charged Count Devices
b) Change Coupled Devices
c) Charge Coupled Devices
d) Charged Compact Disk
Answer: b
Explanation: In dynamic memory, CCD stands for Charge Coupled Devices.

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51. Volatile memory refers to __________
a) The memory whose loosed data is achieved again when power to the memory circuit is removed
b) The memory which looses data when power to the memory circuit is removed
c) The memory which looses data when power to the memory circuit is applied
d) The memory whose loosed data is achieved again when power to the memory circuit is applied

Answer: b
Explanation: Volatile means 'liable to change rapidly' and volatile memory refers to the memory which looses data rapidly when power to the memory circuit is removed. Thus, it looks after it's data as long as it is powered. Non-volatile means 'not volatile' and non-volatile memory refers to the memory which retains the data even if there is a break in the power supply.

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52. Non-volatile memory refers to __________
a) The memory whose loosed data is retained again when power to the memory circuit is removed/applied
b) The memory which looses data when power to the memory circuit is removed
c) The memory which looses data when power to the memory circuit is applied
d) The memory whose loosed data is achieved again when power to the memory circuit is applied
Answer: a
Explanation: Volatile means 'liable to change rapidly' and volatile memory refers to the memory which looses data rapidly when power to the memory circuit is removed. Thus, it looks after it's data as long as it is powered. Non-volatile means 'not volatile' and non-volatile memory refers to the memory which retains the data even if there is a break in the power supply.

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53. The example of non-volatile memory device is __________
a) Magnetic Core Memory
b) Read Only Memory

c) Random Access Memory
d) Both Magnetic Core Memory and Read Only Memory
Answer: d
Explanation: Non-volatile means 'not volatile' and non-volatile memory refers to the memory which retains the data even if there is a break in the power supply. The examples of non-volatile memory devices are Magnetic Core Memory & ROM because both have capability to retain the data.

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54. Based on material used for construction, memory devices are classifieds into ________ categories.
a) 2
b) 3
c) 4
d) 5

Answer: a
Explanation: Based on material used for construction, memory devices are classifieds into two categories, viz., Magnetic and Semiconductor memory. Magnetic recording is the process of storing data magnetically. Hard disk, floppy disk, magnetic tape are examples of magnetic recording process.

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55. Magnetic recording is the process of __________
a) Storing data symmetrically
b) Storing data sequentially
c) Storing data magnetically
d) Both storing data symmetrically and
Answer: c
Explanation: Based on material used for construction, memory devices are classifieds into two categories, viz., Magnetic and Semiconductor memory. Magnetic recording is the process of storing data magnetically. Hard disk, floppy disk, magnetic tape are examples of the magnetic recording process.

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56. Magnetic drum is a storage medium using __________
a) The surface of a jumping magnetic drum
b) The surface of a rotating magnetic drum
c) The surface of a stopped magnetic drum
d) The surface of a moving magnetic drum
Answer: b
Explanation: Magnetic drum is a storage medium using the surface of a rotating magnetic drum which have tendency to hold the data.

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57. Magnetic core is the digital memory in which data is stored magnetically in individual cores operated by __________

a) Up and down select wires

b) Row and column select wires

c) Serial and parallel select wires

d) Up and Serial select wires

Answer: b

Explanation: Magnetic core is the digital memory in which data is stored magnetically in individual cores operated by row and column select wires, with data obtained from sense wire.

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58. By which technology, semiconductor memories are constructed?

a) PLD

b) LSI

c) VLSI

d) Both LSI and VLSI

Answer: d

Explanation: Generally, semiconductor memories are constructed using Large Scale Integration (LSI) or Very Large Scale Integration (VLSI) because these are made up of NMOS, CMOS, BJT, etc.

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59. When two or more devices try to write data in a bus simultaneously, is known as ______________

a) Bus collisions

b) Address multiplexing

c) Address decoding

d) Bus contention

Answer: d

Explanation: Bus contention is an undesirable state of the bus of a computer, in which more than one memory mapped device or the CPU is attempting to place output values onto the bus at once.

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60. A memory is a collection of ____________

a) Unit cells

b) Storage cells

c) Data cells

d) Binary cells

Answer: b

Explanation: A memory is a collection of storage cells with associated circuits needed to transfer information.

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61. To transfer the information from input to output and vice versa, the cells used are ____________

a) Storage cells
b) Data cells
c) Unit cells
d) Both data and unit cells

Answer: a
Explanation: To transfer the information from input to output and vice versa, the cells used are called storage cells. The storage cells stores data in the form of binary information.

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62. The data stored in a group of bits is called ____________
a) Nibble
b) Word
c) Byte
d) Address

Answer: b
Explanation: The data stored in a group of bits is called word. Usually, a word is a group of 16-bits or 2-bytes.

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63. Each word consist of a sequence of ____________
a) Letters
b) Binary numbers
c) Hexadecimal numbers
d) Gray codes
Answer: b
Explanation: Each word consists of a sequence of 0s and 1s (i.e. binary numbers). Usually, a word is a group of 16-bits or 2-bytes.

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64. Each word stored in a memory location is represented by ____________
a) RAM
b) ROM
c) Storage class
d) Address
Answer: d

Explanation: Each word stored in a memory location is represented by address. Usually, a word is a group of 16-bits or 2-bytes.

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65. The group of each 8-bit is called ____________
a) Nibble
b) Flag
c) Byte
d) Word
Answer: c
Explanation: 1 byte = 8-bit, 4-bits = 1 nibble and 16-bits = 1 word.

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66. The capacity of a memory unit is ____________
a) The number of binary input stored
b) The number of words stored
c) The number of bytes stored
d) All of the Mentioned

Answer: c
Explanation: The total number of bytes that can be stored, is the maximum capacity of a memory unit. However, memory unit is the smallest unit of a processor.

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67. The communication between memory and its environment is achieved through

____
a) Control lines
b) Data input/output lines
c) Address selection lines
d) All of the Mentioned
Answer: d
Explanation: Firstly, the data input is needed to transfer the information and it is passed through the address lines and then controlled by control lines. The control lines are responsible for the timing and control of the signals sent and received.

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68. One of the most important specifications on magnetic media is the ____________
a) Polarity reversal rate
b) Tracks per inch
c) Data transfer rate
d) Rotation speed

Answer: c
Explanation: The rate of data transfer depends on the properties of magnetic media.

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